

Vocabulary

Standard Form: A quadratic equation written as $ax^2 + bx + c = 0$, where $a \neq 0$.

Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, a formula that solves any equation written in standard form.

Discriminant: The expression $b^2 - 4ac$, found under the radical, which tells how many real solutions an equation has.

We do together. Solve $2x^2 + 7x - 4 = 0$ using the quadratic formula.

Step 1. Identify the coefficients: $a = 2$, $b = 7$, $c = -4$.

Step 2. Compute the discriminant: $b^2 - 4ac = 7^2 - 4(2)(-4) = 49 + 32 = 81$.

Step 3. Take the square root: $\sqrt{81} = 9$.

Step 4. Substitute into the formula: $x = \frac{-7 \pm 9}{2(2)} = \frac{-7 \pm 9}{4}$.

Step 5. Simplify both solutions: $x = 0.5$ or $x = -4$.

You Try. Solve each. Show every step, then name the answer.

a) Solve $x^2 + 8x + 15 = 0$ using the quadratic formula. $a = 1, b = 8, c = 15$; discriminant $= 64 - 60 = 4 > 0$ (two real roots); $x = \frac{-8 \pm 2}{2}$ so $x = -3$ or $x = -5$.

b) Solve $x^2 - 9x + 20 = 0$ using the quadratic formula. $a = 1, b = -9, c = 20$; discriminant $= 81 - 80 = 1 > 0$ (two real roots); $x = \frac{9 \pm 1}{2}$ so $x = 5$ or $x = 4$.

Summary (fill in): To solve a quadratic equation, first write it in standard form $ax^2 + bx + c = 0$, then substitute a , b , and c into $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The discriminant $b^2 - 4ac$ predicts the outcome: if it is positive there are two real solutions, if it is zero there is exactly one real solution, and if it is negative there are no real solutions.